

Analysis of Point Cloud Domain Gap Effects for 3D Object Detection Evaluation

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Keywords: Point Clouds, Deep Learning, Domain Gap, Object Detection, Simulation.

Abstract: The development of autonomous driving systems heavily relies on high-quality LiDAR data, which is essential for robust object detection and scene understanding. Nevertheless, obtaining a substantial amount of such data for effective training and evaluation of autonomous driving algorithms is a major challenge. To overcome this limitation, recent studies are taking advantage of advancements in realistic simulation engines, such as CARLA, which have provided a breakthrough in generating synthetic LiDAR data that closely resembles real-world scenarios. However, these data are far from being identical to real data. In this study, we address the domain gap between real LiDAR data and synthetic data. We train deep-learning models for object detection using real data. Then, those models are rigorously evaluated using synthetic data generated in CARLA. By quantifying the discrepancies between the model's performance on real and synthetic data, the present study shows that there is indeed a domain gap between the two types of data and does not affect equal to different model architectures. Finally, we propose a method for synthetic data processing to reduce this domain gap. This research contributes to enhancing the use of synthetic data for autonomous driving systems.

1 INTRODUCTION

In the current context of the automotive industry, road safety and autonomous driving have emerged as critical areas of research and development. Accurate and reliable detection of objects in a vehicle's near environment is essential to ensure the safety of passengers, pedestrians, and other road users. One of the key challenges in achieving this level of accuracy and reliability lies in the development of advanced sensor technologies that can provide real-time data about the surrounding environment. As technology advances, LiDAR point clouds have arisen as a promising source of three-dimensional data, offering a detailed representation of the environment around the vehicle (Li

and Ibanez-Guzman, 2020; Li et al., 2021a). These representation presents an opportunity to significantly enhance 3D object detection systems compared to traditional image-based methods (Li et al., 2021b).

Data acquisition is crucial for developing and validating automotive object detection algorithms. For example, Kalra and Paddock (2016) estimate that achieving a 1.09 fatalities per 100 million miles rate with 95% confidence for autonomous vehicles would require 275 million miles. However, obtaining large, diverse, and representative real-world driving datasets can be costly and limiting in terms of time and resources. To address these limitations, the generation of synthetic data is gaining popularity as a practical solution. Simulators such as Dosovitskiy et al. (2017) and Rong et al. (2020) enable the creation of virtual environments that accurately replicate driving conditions, offering the ability to generate substantial volumes of data in a controlled and diverse manner.

Generated data is often used to train deep learning models or to validate automotive functions. However,

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用於 3D 物件偵測評估之點雲域間隙效應分析

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關鍵字：點雲、深度學習、領域差距、物件偵測、模擬。

摘要：自動駕駛系統的發展高度依賴高品質的 LiDAR 資料，這對於穩健的物件偵測和場景理解至關重要。

然而，要獲得大量此類資料以有效訓練和評估自動駕駛演算法是一個重大挑戰。為克服此限制，近期研究正利用逼真模擬引擎（例如 CARLA）的進步，這在生成與真實世界場景高度相似的合成 LiDAR 資料方面取得了突破。然而，這些資料與真實資料仍有顯著差異。在本研究中，我們探討了真實 LiDAR 資料與合成資料之間的領域差距。我們使用真實資料訓練用於物件偵測的深度學習模型。然後，這些模型使用 CARLA 中生成的合成資料進行嚴格評估。透過量化模型在真實資料和合成資料上性能的差異，本研究表明兩種類型的資料之間確實存在領域差距，並且不會影響到不同的模型架構。最後，我們提出了一種合成資料處理方法來縮小此領域差距。

本研究有助於提升合成資料在自動駕駛系統中的應用。

1 緒論

在當前的汽車產業中，道路安全和自動駕駛已成為研究和開發的關鍵領域。準確可靠地偵測車輛周圍環境中的物件，對於確保乘客、行人和其他道路使用者的安全至關重要。實現這種準確性和可靠性的主要挑戰之一，在於開發先進的感測器技術，這些技術可以提供有關周圍環境的即時數據。隨著技術的進步，光達點雲已成為一種有前景的三維數據來源，可提供車輛周圍環境的詳細表示（Li

org/0000-0001-9879-0992 和 Ibanez-Guzman，2020 年；Li 等人，2021a）。與傳統的基於影像的方法相比，這些表示法為顯著增強 3D 物件偵測系統提供了機會（Li 等人，2021b）。

資料擷取對於開發和驗證汽車物件偵測演算法至關重要。例如，Kalra 和 Paddock (2016) 估計，若要讓自動駕駛車輛以 95% 的信心達到每 1 億英里 1.09 死亡人數的比例，將需要行駛 2.75 億英里。然而，獲取大量、多樣化且具代表性的真實世界駕駛資料集可能成本高昂，並在時間和資源方面受到限制。為了解決這些限制，合成資料的生成正日益普及，成為一種實用的解決方案。Dosovitskiy 等人 (2017) 和 Rong 等人 (2020) 的模擬器能夠建立精確複製駕駛條件的虛擬環境，提供以受控且多樣化的方式生成大量資料的能力。

生成的資料通常用於訓練深度學習模型或驗證汽車功能。然而，

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Studies (Dworak et al., 2019; Huch et al., 2023) reveal a domain gap between real and synthetic data, which can negatively affect the results. The domain gap refers to the mismatch or disparity between the data distribution. This paper examines the domain gap in point cloud-based object detection models trained with real data and presents a method to mitigate it.

In the present study, we have designed a workflow as shown in Figure 1 to improve the understanding of the domain gap between synthetic and real point cloud data by performing the following contributions:

- **Synthetic Data Generation Pipeline.** We propose a synthetic data generation pipeline to generate data in a standardized format, which can be easily adapted to different sensor setups. We used this pipeline to generate our own synthetic dataset: <https://datasets.vicomtech.org/di21-carla-point-cloud-data/carla-point-cloud-data.md>.
- **Analysis of the Generalization of Models Against Synthetic Data.** We evaluate different architectures of LiDAR-based 3D object detection models with the generated synthetic dataset. We then analyze the ability of each model to adapt to this domain.
- **Investigation of the Domain Gap Between Real and Synthetic Data.** We compare the generated synthetic point clouds with point clouds of a dataset with real data to identify the differences between both domains.
- **Domain Gap Reduction.** We propose a method to reduce the domain gap between real and synthetic data. This method is able to reduce the precision difference of the models between both domains.

The rest of this paper is organized into five sections. The first section presents the state of the art of point cloud datasets and deep learning models, especially emphasizing the existing knowledge gap regarding the response of models to synthetic data. The second section describes the methodology used during this research. The third section details the experiments done during the investigation. In the fourth section, an exhaustive analysis of the results can be found. Finally, the last section compiles the conclusions and future work of the research.

2 RELATED WORK

The development of autonomous driving has relied heavily on the availability of high-quality datasets

for training and validating autonomous driving algorithms. Among the most popular datasets are KITTI (Liao et al., 2021), Waymo (Sun et al., 2019), and nuScenes (Caesar et al., 2019). However, the datasets used in autonomous driving present some challenges, such as data limitations, particularly in acquiring data for rare or extreme driving scenarios that are difficult to obtain.

Given the considerable expense and practical difficulties associated with obtaining only real data to meet the necessary quantity, variability, and diversity, synthetic data are widely utilized as supplementary resources (Qiao and Zulkernine, 2023; Wang et al., 2019; Inan et al., 2023). These data are generated to cover hazardous or unusual situations that may not be easily found in real data and avoid the process and cost associated with obtaining and annotating real data. For the generation of these data, simulation environments such as CARLA (Dosovitskiy et al., 2017) or LGSVL (Rong et al., 2020) are used and even datasets exclusively with synthetic data have been published (Kloukiniotis et al., 2022; Sekkat et al., 2022; Xu et al., 2022).

Synthetic data are often used to study functions such as object detection. Object detection in point clouds is a constantly evolving area of research in the fields of machine perception, computer vision, and robotics (Li et al., 2021b). This task is very challenging due to the complexity and variability of point clouds, which are sparse and unordered and may contain noise, occlusions, and objects with very different shapes and sizes. In recent years, several deep learning models have been proposed for the detection of objects in point clouds (Lang et al., 2018; Zhu et al., 2020; Yin et al., 2020; Bai et al., 2022). However, these models are still far from achieving the results needed for a fully autonomous vehicle as described by Martínez-Díaz and Soriguera (2018). However, these models are usually trained on public datasets where capturing the full spectrum of real-world edge scenarios is often unfeasible, resulting in a knowledge gap regarding how the trained models will perform in critical edge scenarios. Consequently, the use of synthetic data has emerged as a pivotal strategy to supplement the available datasets.

Although different research proves the effectiveness of synthetic data, different works (Dworak et al., 2019; Huch et al., 2023) demonstrate that the impact of a domain gap may diverge across deep learning models depending on their architecture. This is especially accentuated when using synthetic data. Furthermore, the state-of-the-art LiDAR-based 3D detection models are not analyzed in terms of generalization or sensitivity to this domain gap.

研究 (Dworak et al., 2019; Huch et al., 2023) 揭示了真實資料與合成資料之間的領域差異，這可能會對結果產生負面影響。領域差異指的是資料分佈之間的不匹配或差異。本文探討了使用真實資料訓練的點雲物件偵測模型中的領域差異，並提出了一種減輕該差異的方法。

在本研究中，我們設計了一個如圖 1 所示的工作流程，透過以下貢獻來增進對合成與真實點雲資料之間領域差距的理解：

- 合成資料生成管線。我們提出了一個合成資料生成管線，以標準化格式生成資料，該格式可以輕鬆適應不同的感測器設置。我們使用此管線生成我們自己的合成資料集：
<https://datasets.vicomtech.org/di21-carla-point-cloud-data/>
`carla-point-cloud-data.md`。
- 模型對合成資料泛化能力的分析。我們使用生成的合成資料集評估了基於 LiDAR 的 3D 物件偵測模型的不同架構。然後，我們分析了每個模型適應此領域的能力。
- 真實與合成資料之間的領域差距調查。我們將生成的合成點雲與真實資料集的點雲進行比較，以識別兩個領域之間的差異。
- 縮小領域差距。我們提出了一種方法來縮小真實與合成資料之間的領域差距。此方法能夠減少模型在兩個領域之間的精確度差異。

本文其餘部分分為五個章節。第一節介紹了點雲資料集和深度學習模型的最新技術，特別強調了模型對合成資料的回應方面現有的知識差距。第二節描述了本研究中使用的方法。第三節詳細介紹了調查期間進行的實驗。第四節可以找到對結果的詳盡分析。最後，最後一節彙編了本研究的結論和未來工作。

用於訓練和驗證自動駕駛演算法。其中最受歡迎的資料集包括 KITTI (Liao et al., 2021)、Waymo (Sun et al., 2019) 和 nuScenes (Caesar et al., 2019)。然而，自動駕駛中使用的資料集存在一些挑戰，例如資料限制，特別是在獲取難以獲得的稀有或極端駕駛情境資料方面。

鑑於僅獲取真實資料以滿足必要的數量、變異性和多樣性所需的大量費用和實際困難，合成資料被廣泛用作補充資源 (Qiao and Zulkernine, 2023; Wang et al., 2019; Inan et al., 2023)。這些資料的生成旨在涵蓋真實資料中可能不易發現的危險或異常情況，並避免與獲取和註釋真實資料相關的過程和成本。為了生成這些資料，使用了 CARLA (Dosovitskiy et al., 2017) 或 LGSVL (Rong et al., 2020) 等模擬環境，甚至已經發布了僅包含合成資料的資料集 (Klounikiotis et al., 2022; Sekkat et al., 2022; Xu et al., 2022)。

合成資料通常用於研究諸如物件偵測等功能。點雲中的物件偵測是機器感知、電腦視覺和機器人學領域中不斷發展的研究領域 (Li et al., 2021b)。這項任務極具挑戰性，因為點雲的複雜性和可變性，它們稀疏且無序，並且可能包含雜訊、遮擋以及形狀和大小差異很大的物件。近年來，已經提出了幾種深度學習模型用於點雲中的物件偵測 (Lang et al., 2018; Zhu et al., 2020; Yin et al., 2020; Bai et al., 2022)。然而，這些模型距離實現 Martínez-Díaz 和 Soriguera (2018) 所描述的完全自動駕駛車輛所需的結果仍有很長的距離。然而，這些模型通常在公共資料集上進行訓練，在這些資料集中，捕捉真實世界邊緣情境的全部範圍通常是不可行的，導致在關鍵邊緣情境中訓練模型的表現存在知識差距。因此，使用合成資料已成為補充可用資料集的關鍵策略。

儘管不同的研究證明了合成資料的有效性，但不同的著作 (Dworak 等人，2019 年；Huch 等人，2023 年) 表明，領域差距的影響可能會因深度學習模型的架構而異。當使用合成資料時，這種差異尤其明顯。此外，目前最先進的基於光達的 3D 偵測模型在泛化能力或對此領域差距的敏感度方面尚未進行分析。

2 相關工作

自動駕駛的發展在很大程度上依賴於高品質資料集的可用性

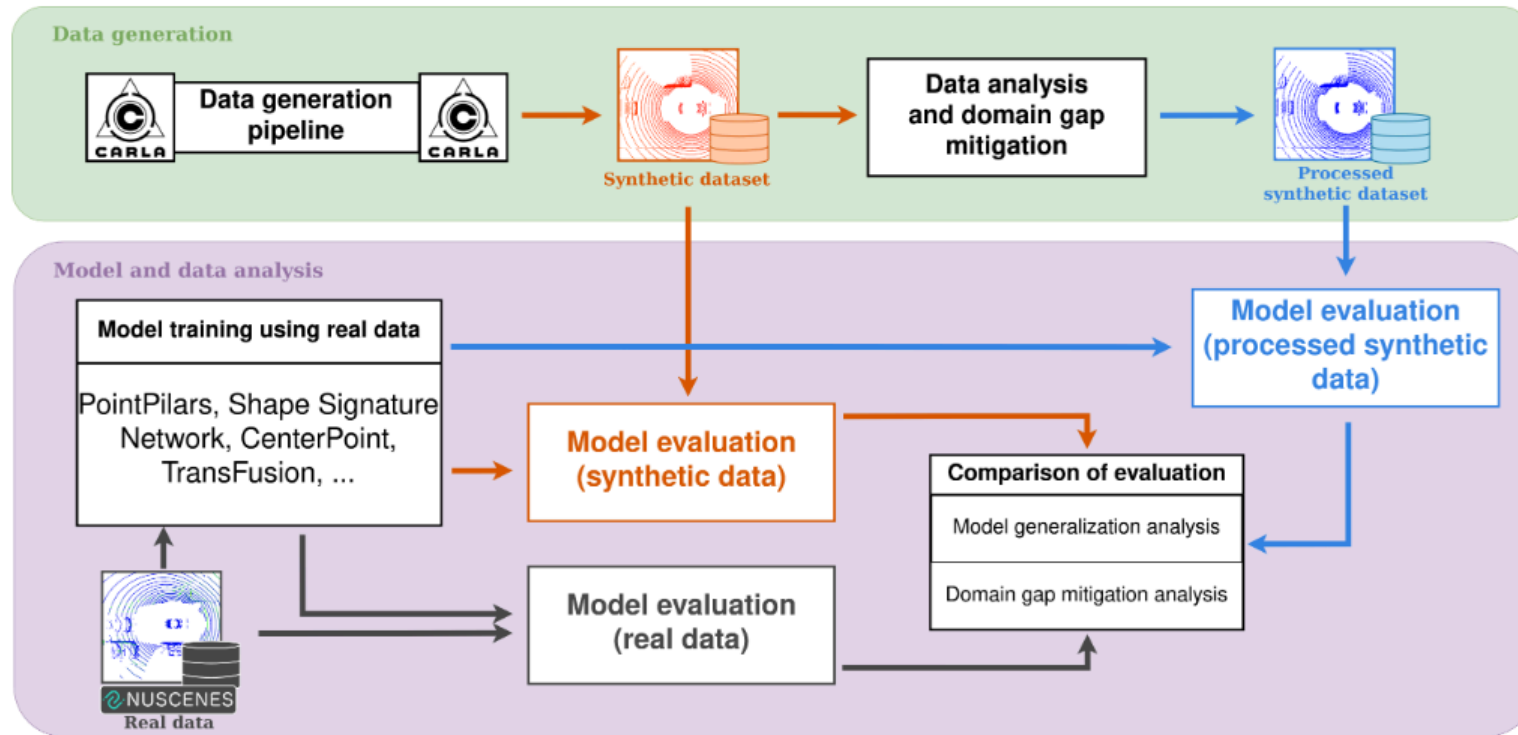


Figure 1: Methodology of the research. First, we generate a synthetic dataset, analyze it, and propose a domain gap mitigation strategy. Then, we train different architectures of LiDAR-based 3D object detection models and evaluate their performance with real data, synthetic data, and processed synthetic data.

3 METHODOLOGY

The methodology proposed in this research is divided into two parts, as can be seen in Figure 1. The first part is focused on data generation and processing, and the second one is about model training and evaluation. In the first part, we first generate a synthetic dataset using the CARLA simulator, thenceforth we study the peculiarities of these data and propose a strategy to mitigate the potential domain gap. In the second part, we first train state-of-the-art LiDAR-based 3D object detection models with different architectures. Subsequently, we evaluate these models with a real-world dataset (the nuScenes dataset in this case), the synthetic dataset, and the processed synthetic dataset. Finally, the results of the evaluations for the different domains are compared and analyzed.

3.1 Data Generation Pipeline

In this work, we design a pipeline for the creation of synthetic datasets using the CARLA simulator (Dosovitskiy et al., 2017). This dataset will later be used to evaluate object detection models in point clouds. We use CARLA due to its extensive usage in state-of-the-art research (Kloukiniotis et al., 2022; Sekkat et al., 2022; Xu et al., 2022) and its versatile capabilities, but other alternatives could be used as well.

3.1.1 Data Acquisition

To generate point clouds using the CARLA simulator we use two virtual sensors attached to a vehicle of

the simulator. First, we use a LiDAR sensor to generate point clouds. Second, we use a semantic LiDAR, which does not represent a real sensor but provides the type of object the simulated LiDAR rays collide with. Both LiDARs are needed to generate the data and the corresponding automatic annotations. The data generation process is conducted as follows.

First, before data generation, we define the sensor information of our setup in the standardized OpenLABEL¹ format. This format defines how information on the extrinsic and intrinsic parameters of the sensors of the vehicle must be indicated. The flexibility of this format allows easily defining different sensor setups to be simulated. Once we have the sensor information defined in an OpenLabel format file, the CARLA map where the scene will take place is selected and loaded. Then, we load the ego-vehicle and attach the sensors specified in the file. After loading the map, the sensors, and the vehicle, several actors are generated at different points of the scene, these actors are selected from a distribution of cars, trucks, vans, buses, bikes, motorcycles, and pedestrians. After the loading of different actors, all of them are activated in auto-pilot mode, enabling independent movement in compliance with traffic regulations.

Before starting to generate and annotate data we define a margin of 200 frames (20 seconds) for each vehicle to reach the appropriate speed for the position they are in, as when they are loaded, they are stationary. Within frame, a point cloud is stored. Then, we use the semantic LiDAR to automatically generate an-

¹<https://www.asam.net/standards/detail/openlabel/>

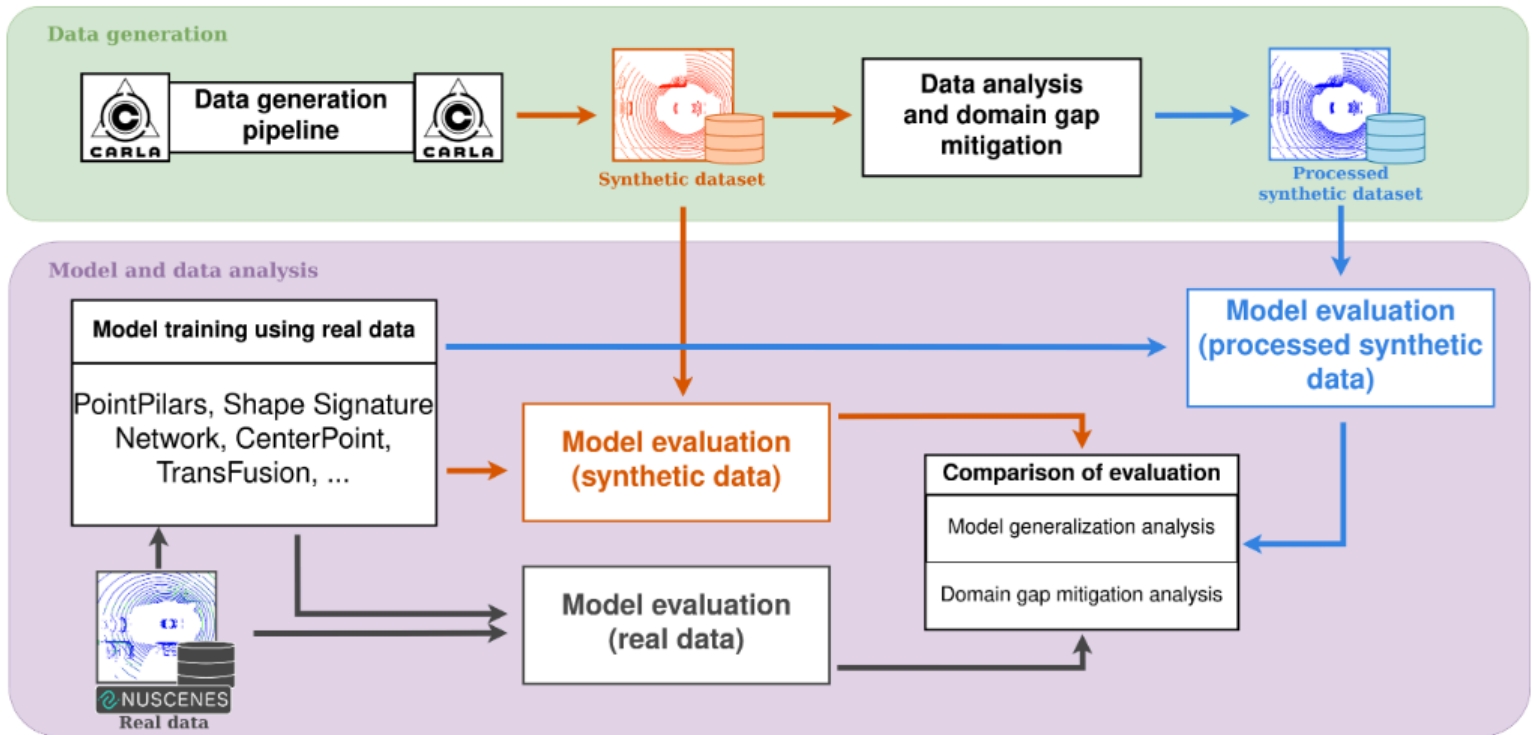


圖 1：研究方法。首先，我們生成一個合成資料集，對其進行分析，並提出一個領域差距緩解策略。然後，我們訓練不同架構的基於 LiDAR 的 3D 物件偵測模型，並使用真實資料、合成資料和處理過的合成資料來評估它們的效能。

3 方法論

本研究提出的方法論分為兩個部分，如圖 1 所示。第一部分著重於資料生成與處理，第二部分則關於模型訓練與評估。在第一部分中，我們首先使用 CARLA 模擬器生成合成資料集，然後研究這些資料的特性並提出策略以減輕潛在的領域差異。在第二部分中，我們首先訓練具有不同架構的最新 LiDAR 式 3D 物件偵測模型。隨後，我們使用真實世界資料集（在本例中為 nuScenes 資料集）、合成資料集和經過處理的合成資料集來評估這些模型。最後，比較並分析不同領域的評估結果。

3.1 資料生成管線

在這項工作中，我們設計了一個使用 CARLA 模擬器（Dosovitskiy 等人，2017 年）建立合成資料集的管線。此資料集將用於評估點雲中的物件偵測模型。我們使用 CARLA 是因為它在最新研究中廣泛使用（Kloutiniotis 等人，2022 年；Sekkat 等人，2022 年；Xu 等人，2022 年）及其多功能性，但也可以使用其他替代方案。

3.1.1 資料擷取

為了使用 CARLA 模擬器生成點雲，我們使用了連接到車輛的兩個虛擬感測器，其中一個是

模擬器。首先，我們使用光達感測器來生成點雲。其次，我們使用語義光達，它不代表真實感測器，而是提供模擬光達射線碰撞到的物件類型。這兩種光達都需要用於生成資料和相應的自動註釋。資料生成過程如下。

首先，在資料生成之前，我們以標準化的 OpenLABEL 格式定義了我們設定的感測器資訊。此格式定義了車輛感測器的外部和內部參數資訊必須如何表示。此格式的靈活性允許輕鬆定義要模擬的不同感測器設定。一旦我們在 OpenLabel 格式檔案中定義了感測器資訊，就會選擇並載入場景將發生的 CARLA 地圖。然後，我們載入自我車輛並連接檔案中指定的感測器。載入地圖、感測器和車輛後，會在場景的不同點生成多個參與者，這些參與者是從汽車、卡車、廂型車、公車、自行車、摩托車和行人的分佈中選擇的。載入不同的參與者後，所有參與者都以自動駕駛模式啟動，從而能夠根據交通法規獨立移動。

在開始生成和註釋資料之前，我們為每輛車定義了 200 幀（20 秒）的裕度，以使其達到適合其所在位置的速度，因為它們在載入時是靜止的。在幀內，會儲存一個點雲。然後，我們使用語義 LiDAR 自動生成一個

¹<https://www.asam.net/standards/detail/openlabel/>

notations for the actors appearing in the point cloud. Generated data can be seen in Figure 2.

3.1.2 Data Structure

The generated dataset can be categorized into two primary components: point clouds and their corresponding annotations, with the annotations being stored in OpenLABEL format. Each scene is defined by an OpenLABEL file, containing the timestamp of every frame. Each frame encapsulates the following data:

- Transformation matrix: This field holds a matrix detailing the vehicle's position and orientation in relation to the ego vehicle's initial position within the frame.
- Objects: This section comprises annotations for the objects depicted in the frame, each consisting of: 3D bounding box, velocity vector, acceleration vector, velocity, acceleration and class.

3.2 Data Analysis and Domain Gap Mitigation

Once we generate the data, we compare the synthetic point clouds with point clouds of a real-world dataset (more specifically, the nuScenes dataset). We identify the following differences:

1. All the lasers used by a simulated LiDAR always obtain information. This does not happen with a real LiDAR, because either by the surface or the material with which the laser collides or by the weather, many of the points are lost. However, this does not happen in a simulated environment.
2. The points form perfect circles in synthetic point clouds. This is a result of the simulation environment. Real LiDAR cannot obtain the same accuracy as sensors used in simulators, due to sensor errors, surface irregularities, weather, or even motion distortion. This is why real point clouds do not have perfect rings like synthetic clouds.
3. Finally, another factor that causes these domains to be different is the intensity value of the points. Intensity calculation of CARLA lacks realism as it solely relies on point distance, it does not take into account the material with which the laser collides or the angle of incidence.

These differences can be seen in Figure 3, which illustrates variations between a point cloud from the nuScenes dataset and a synthetic dataset. The color of the points varies based on the intensity of the points. Based on the detected data disparities, we propose to process the virtually generated clouds to mitigate the difference between the two domains as follows.

1. To solve the problem of point loss in the real environment, some points are randomly removed from the cloud.
2. To solve the problem of perfect virtual point clouds, in comparison with the inherent noise obtained from the measurements of real LiDAR, we add a random noise distribution to the points.
3. Finally, to deal with the problem of intensity values, we propose to change all intensity values to a common value.

3.3 Model and Data Analysis

To study the differences between real and synthetic data, and the performance of object detection models on real and synthetic point clouds. We train different architectures of these models, along with models having various configurations in terms of point cloud accumulation. In this way, a more exhaustive study of the domain gap between real and synthetic data can be performed. We train the models with real data and then evaluate them with real data as well as with synthetic and processed synthetic data.

4 EXPERIMENTS

In the upcoming section, we delve into the experimental phase of our study, where our first objective is to generate a synthetic dataset. We list the parameters used in data processing and summarize the obtained dataset. We also present the LiDAR-based 3D models we utilized, as well as their configuration.

4.1 Synthetic Data

The generated dataset consists of 9,600 point clouds, which correspond to 96 scenes with 100 point clouds each, at a frame rate of 0.1 seconds. Each point cloud is labeled with information about the objects present in it, encompassing 6 different classes: car, pedestrian, bicycle, motorcycle, bus, and truck. Detailed quantitative information can be found in Table 1.

4.1.1 Data Generation

For the generation of these data, four maps from the CARLA simulation environment are used to give more variety to the dataset scenes these maps have rural and urban environments. In addition, the default weather and time settings are used, because, in the simulator, the LiDAR sensor is not affected by these changes. In each scene, 30 cars, 10 trucks,

點雲中出現的演員的標記。
生成的資料如圖 2 所示。

3.1.2 資料結構

生成的資料集可分為兩個主要部分：點雲及其對應的註釋，註釋以 OpenLABEL 格式儲存。每個場景都由一個 OpenLABEL 檔案定義，其中包含每個影格的時間戳記。每個影格都包含以下資料：

- 轉換矩陣：此欄位包含一個矩陣，詳細說明車輛相對於自我車輛在影格中的初始位置的位置和方向。

- 物件：此部分包含影格中描繪之物件的註釋，每個註釋都包含：3D 邊界框、速度向量、加速度向量、速度、加速度和類別。

3.2 資料分析與領域差距緩解

產生資料後，我們會將合成點雲與真實世界資料集（更具體地說，是 nuScenes 資料集）的點雲進行比較。我們發現以下差異：1. 模擬 LiDAR 使用的所有雷射始終都能獲取資訊。真實 LiDAR 不會發生這種情況，因為無論是雷射碰撞的表面或材料，還是天氣，許多點都會遺失。然而，在模擬環境中不會發生這種情況。

2. 點在合成點雲中形成完美的圓圈。這是模擬環境的結果。真實 LiDAR 無法獲得與模擬器中使用的感測器相同的準確度，因為感測器誤差、表面不規則、天氣，甚至運動失真。這就是為什麼真實點雲不像合成點雲那樣具有完美的環。
3. 最後，導致這些領域不同的另一個因素是點的強度值。

CARLA 的強度計算缺乏真實性，因為它僅依賴點距離，沒有考慮雷射碰撞的材料或入射角。

這些差異可見於圖 3，該圖說明了 nuScenes 資料集中的點雲與合成資料集之間的差異。點的顏色根據點的強度而變化。根據檢測到的資料差異，我們建議如下處理虛擬生成的點雲，以減輕兩個領域之間的差異。

1. 為了解決真實環境中點損失的問題，從點雲中隨機移除一些點。
2. 為了解決完美虛擬點雲的問題，與從真實光達測量中獲得的固有雜訊相比，我們為點添加了隨機雜訊分佈。

3. 最後，為了解決強度值的問題，我們建議將所有強度值更改為一個共同值。

3.3 模型與資料分析

為了研究真實與合成資料之間的差異，以及物件偵測模型在真實與合成點雲上的效能。我們訓練這些模型的不同架構，以及在點雲累積方面具有各種配置的模型。透過這種方式，可以對真實與合成資料之間的領域差距進行更詳盡的研究。我們使用真實資料訓練模型，然後使用真實資料以及合成和處理過的合成資料進行評估。

4 實驗

在接下來的章節中，我們將深入探討研究的實驗階段，其中我們的首要目標是生成合成資料集。我們列出了資料處理中使用的參數，並總結了所獲得的資料集。我們還介紹了我們使用的基於光達的 3D 模型及其配置。

4.1 合成資料

生成的資料集包含 9,600 個點雲，對應 96 個場景，每個場景有 100 個點雲，幀率為 0.1 秒。每個點雲都標記了其中存在的物件資訊，涵蓋 6 種不同的類別：汽車、行人、自行車、摩托車、公車和卡車。詳細的量化資訊請參閱表 1。

4.1.1 資料生成

為了生成這些資料，我們使用了 CARLA 模擬環境中的四張地圖，以增加資料集場景的多樣性，這些地圖包含鄉村和城市環境。此外，我們使用了預設的天氣和時間設定，因為在模擬器中，光達感測器不受這些變化的影響。在每個場景中，有 30 輛汽車、10 輛卡車、

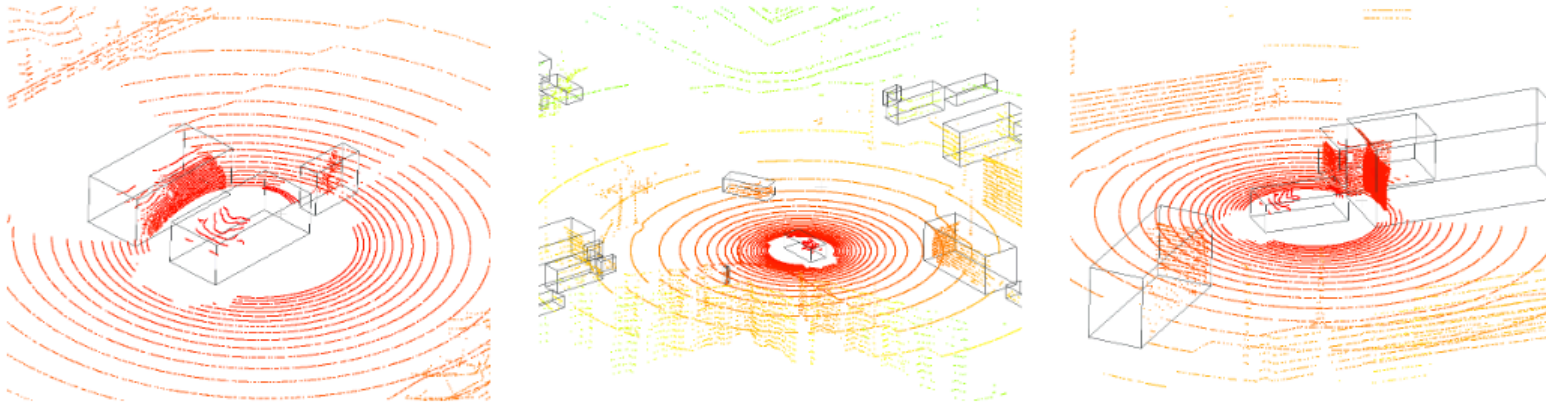


Figure 2: Example of generated point clouds. Each image is a point cloud of a different scene.

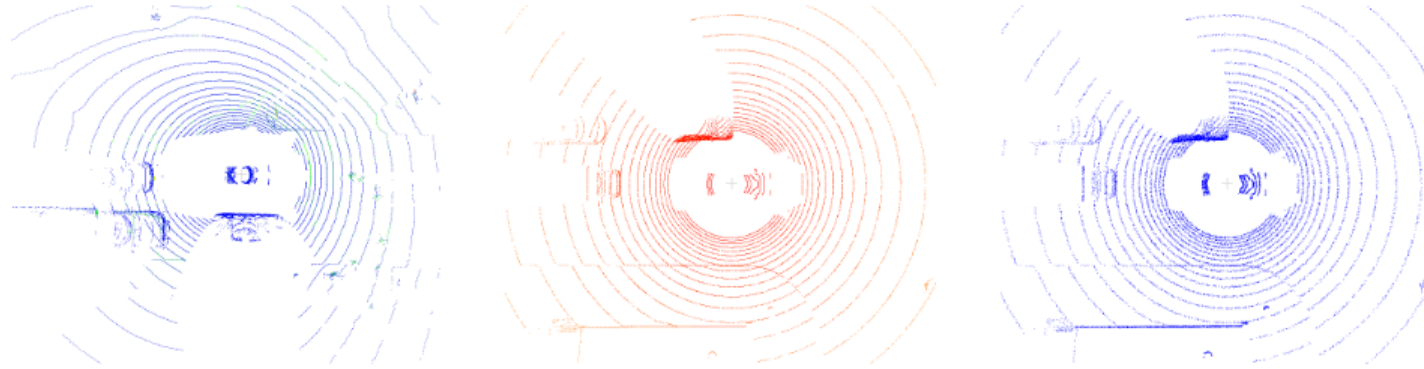


Figure 3: Comparison between point clouds of different domains. In the first column a point cloud of the nuScenes dataset, in the second column a point cloud of the synthetic dataset and in the third column the point cloud of the synthetic dataset after applying our processing on it.

Table 1: Number of annotations of the synthetic dataset.

Object Class	Number of annotations
car	48,878
truck	26,273
bus	13,097
pedestrian	29,373
motorcycle	20,526
bicycle	18,177
total	156,324

10 vans, 10 buses, 20 bikes, 20 motorcycles and 60 pedestrians are loaded. We employ this distribution in an attempt to simulate the distribution of the dataset against which the models are evaluated, the nuScenes dataset (Caesar et al., 2019).

4.1.2 Synthetic Data Processing

The values for the reduction and translation of the points (Section 3.2) are selected trying to obtain clouds as close to reality as possible. We empirically found the optimal percentage of points to remove is 10% and the optimal value for the translation is a range of $[-0.05, 0.05]$ meters in each axis. As for the intensity value, we decide to take the value 1 as it is one of the most common in real point clouds since the car is surrounded by poorly reflecting objects such as the road. Note that these changes do not completely eliminate the domain gap, but they do reduce it. In Figure 3 it can be seen a comparison of the change of the cloud before and after applying the processing.

4.2 Model Configuration and Training

For our experiments, we employ the MMDetection3D library (Contributors, 2020) due to its rich assortment of point cloud object detection models. We use the nuScenes (Caesar et al., 2019) dataset. In this research, we employ two categories of models of the state-of-the-art with different architectures and properties. The first category includes PointPillars (Lang et al., 2018) and Shape Signature Network (Zhu et al., 2020), which convert the point cloud into an intermediate representation like voxels or pillars. The second category comprises CenterPoint (Yin et al., 2020) and TransFusion (Bai et al., 2022) (LiDAR only), which not only transform the point cloud into an intermediate representation but also incorporate the top-view of the point cloud. To assess the impact of point cloud accumulation, we conduct model training in two distinct configurations: one where no sweeps² are incorporated (denoted as "0 sweeps"), and another where 10 sweeps are integrated into the training process. Furthermore, due to the limitations of the CARLA simulator, only 6 classes are considered for this study (car, truck, bus, pedestrian, motorcycle, bicycle) even though nuScenes contains 10 classes. The models are trained replicating the state-of-the-art configuration, we use the usual state-of-the-art 3D object detection

²Sweeps in the nuScenes benchmark refer to the intermediate frames without annotations, while samples refer to those key frames with annotations. Sweeps are used to accumulate point clouds and thus make the main cloud denser.

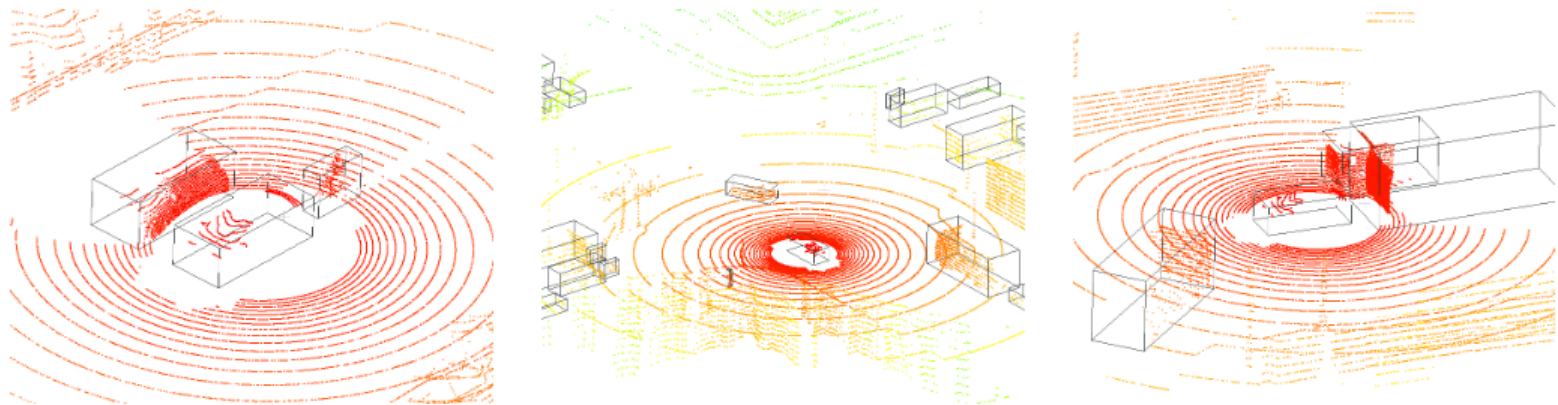


圖 2：生成的點雲範例。每個影像都是不同場景的點雲。

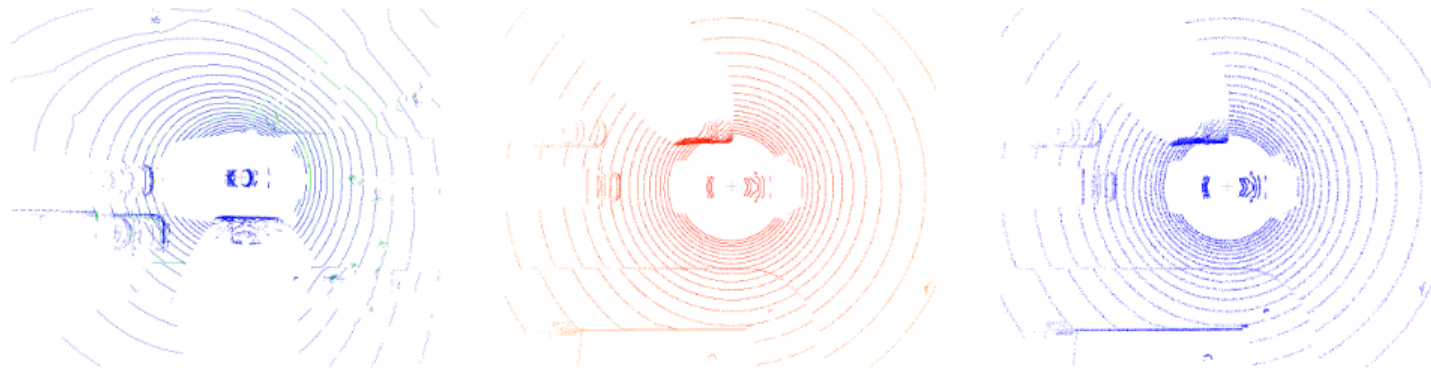


圖 3：不同領域點雲的比較。第一欄是 nuScenes 資料集的點雲，第二欄是合成資料集的點雲，第三欄是經過我們處理後的合成資料集的點雲。

表 1：合成資料集的註釋數量。

物件	類別	註釋數量	汽車 48,878	卡車 26,273	公車 13,097	行人 29,373	機車 20,526	自行車 18,177	總計 156,324
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已載入 10 輛廂型車、10 輛巴士、20 輛自行車、20 輛摩托車和 60 名行人。我們採用此分佈，試圖模擬用於評估模型的資料集（nuScenes 資料集，Caesar 等人，2019 年）的分佈。

4.1.2 合成資料處理

點的縮減和轉換值（第 3.2 節）的選擇，是為了盡可能獲得接近真實的點雲。我們憑經驗發現，移除點的最佳百分比為 10%，而轉換的最佳值是每個軸上 [-0.05, 0.05] 公尺的範圍。至於強度值，我們決定取值 1，因為這是真實點雲中最常見的值之一，因為汽車周圍環繞著反射性差的物體，例如道路。請注意，這些變化並不能完全消除領域差距，但確實減少了差距。圖 3 顯示了應用處理前後點雲變化的比較。

4.2 模型配置與訓練

在我們的實驗中，我們採用 MMDetection3D 函式庫 (Contributors, 2020)，因為它擁有豐富的點雲物件偵測模型。我們使用 nuScenes (Caesar et al., 2019) 資料集。在這項研究中，我們採用了兩種具有不同架構和屬性的最先進模型類別。第一類包括 PointPillars (Lang et al., 2018) 和 Shape Signature Network (Zhu et al., 2020)，它們將點雲轉換為體素或柱狀等中間表示。第二類包括 CenterPoint (Yin et al., 2020) 和 TransFusion (Bai et al., 2022) (僅限 LiDAR)，它們不僅將點雲轉換為中間表示，還結合了點雲的俯視圖。為了評估點雲累積的影響，我們在兩種不同的配置下進行模型訓練：一種是不包含任何掃描 (表示為「0 掃描」)，另一種是將 10 次掃描整合到訓練過程中。此外，由於 CARLA 模擬器的限制，儘管 nuScenes 包含 10 個類別，但本研究僅考慮 6 個類別 (汽車、卡車、公車、行人、摩托車、自行車)。模型按照最先進的配置進行訓練，我們使用通常的最先進 3D 物件偵測。

nuScenes 基準測試中的掃描是指沒有註釋的中間幀，而樣本是指帶有註釋的關鍵幀。掃描用於累積點雲，從而使主點雲更密集。

Table 2: Comparison of model evaluation (average precision) across 0 sweep models using three types of data: real data, synthetic data, and processed synthetic data.

Model	PointPillars (Lang et al., 2018)			SSN (Zhu et al., 2020)			CenterPoint (Yin et al., 2020)			TransFusion (Bai et al., 2022)		
Dataset	Real	Synth	P. synth	Real	Synth	P. synth	Real	Synth	P. synth	Real	Synth	P. synth
car	0.733	0.593	0.605	0.722	0.656	0.669	0.741	0.125	0.588	0.806	0.500	0.771
truck	0.306	0.584	0.582	0.372	0.633	0.630	0.399	0.007	0.428	0.426	0.500	0.831
bus	0.395	0.280	0.258	0.435	0.304	0.293	0.570	0.000	0.216	0.696	0.000	0.271
pedestrian	0.569	0.537	0.620	0.481	0.592	0.578	0.654	0.053	0.312	0.783	0.000	0.983
motorcycle	0.232	0.717	0.708	0.239	0.676	0.677	0.317	0.000	0.024	0.531	0.000	0.057
bicycle	0.034	0.432	0.428	0.027	0.528	0.502	0.180	0.053	0.119	0.285	0.079	0.351
Mean	0.378	0.524	0.533	0.379	0.565	0.558	0.477	0.040	0.281	0.588	0.180	0.544

Table 3: Comparison of model evaluation (average precision) across 10 sweep models using three types of data: real data, synthetic data, and processed synthetic data.

Model	PointPillars (Lang et al., 2018)			SSN (Zhu et al., 2020)			CenterPoint (Yin et al., 2020)			TransFusion (Bai et al., 2022)		
Dataset	Real	Synth	P. synth	Real	Synth	P. synth	Real	Synth	P. synth	Real	Synth	P. synth
car	0.803	0.662	0.673	0.816	0.615	0.636	0.821	0.331	0.563	0.871	0.000	0.770
truck	0.362	0.522	0.515	0.464	0.564	0.584	0.484	0.320	0.590	0.387	0.000	0.693
bus	0.426	0.194	0.180	0.561	0.316	0.309	0.627	0.037	0.247	0.734	0.000	0.246
pedestrian	0.742	0.873	0.841	0.665	0.769	0.732	0.748	0.455	0.949	0.866	0.000	0.974
motorcycle	0.332	0.540	0.533	0.427	0.640	0.626	0.433	0.000	0.054	0.620	0.000	0.000
bicycle	0.094	0.333	0.337	0.158	0.721	0.740	0.302	0.094	0.204	0.294	0.081	0.311
Mean	0.460	0.521	0.513	0.515	0.604	0.604	0.569	0.206	0.434	0.629	0.013	0.499

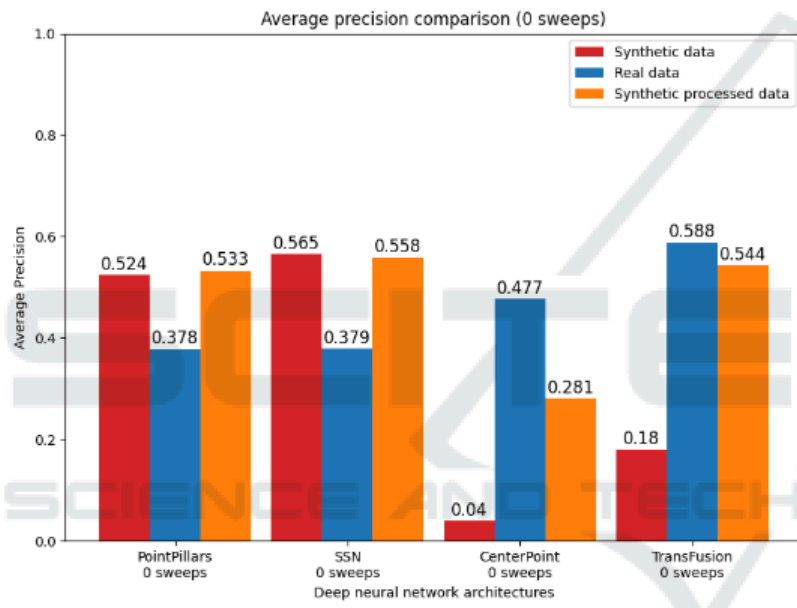


表 2：使用三種類型資料（真實資料、合成資料和處理過的合成資料）對 0 掃描模型進行模型評估（平均精確度）的比較。

模型	PointPillars (Lang et al., 2018)	SSN (Zhu et al., 2020)	CenterPoint (Yin et al., 2020)	TransFusion (Bai et al., 2022)	資料集	真實	合成	處理過的合成
合成	0.500	0.771	0.306	0.584	0.582	0.372	0.633	0.630
真實	0.216	0.696	0.000	0.271	行人	0.569	0.537	0.620
處理過的合成	0.317	0.000	0.024	0.531	0.000	0.057	自行車	0.034
0.379	0.565	0.558	0.477	0.040	0.281	0.588	0.180	0.544

表 3：使用三種類型資料（真實資料、合成資料和處理過的合成資料）對 10 掃描模型進行模型評估（平均精確度）的比較。

模型	PointPillars (Lang et al., 2018)	SSN (Zhu et al., 2020)	CenterPoint (Yin et al., 2020)	TransFusion (Bai et al., 2022)	資料集	Real	Synth	P. synth
Real	0.362	0.522	0.515	0.464	0.564	0.584	0.484	0.320
Synth	0.000	0.246	行人	0.742	0.873	0.841	0.665	0.769
P. synth	0.000	0.054	0.620	0.000	0.000	自行車	0.094	0.333
0.604	0.604	0.569	0.206	0.434	0.629	0.013	0.499	

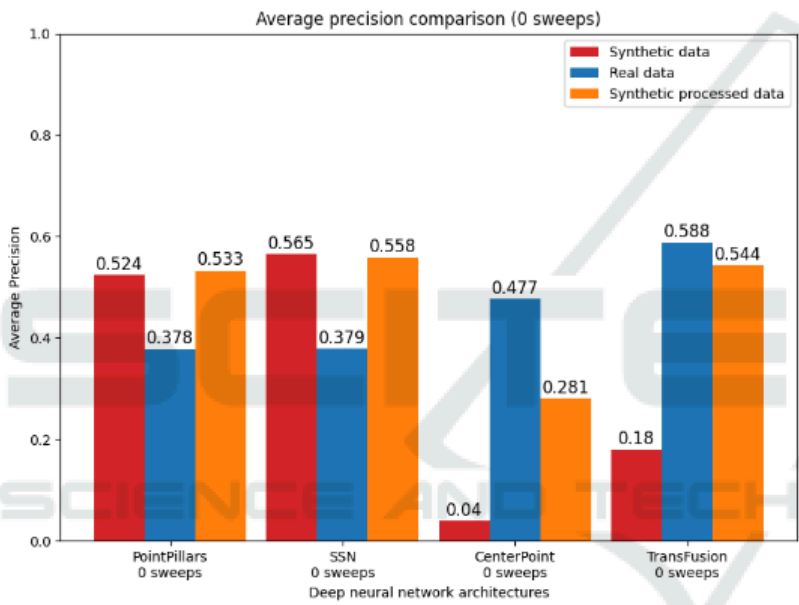


圖 4：每個資料類型的平均精確度值：合成（紅色）、真實（藍色）和合成處理（橙色），每個模型都沒有點雲累積。評估指標。我們主要關注平均精確度 (AP)。模型評估結果可在表 2 和表 3 中找到。

4.3 結果分析

訓練好的模型會使用真實、合成和經過處理的合成資料進行評估。結果如表 2 (0 次掃描) 和表 3 (10 次掃描) 所示。值得注意的是，PointPillars 和 SSN 在合成資料上表現出卓越的性能，而 CenterPoint 和 TransFusion 在真實資料上表現更好。我們認為 CenterPoint 和 TransFusion 準確度下降的原因可能歸因於使用了從點雲頂視圖獲得的熱圖，這種表示方式似乎更容易受到領域差距效應的影響。儘管前兩個模型的準確度差異很小

但對於其他兩個模型來說，差異如此之大，這意味著某些模型無法正確泛化，並且合成資料和真實資料之間確實存在顯著的領域差距。

在將我們提出的處理應用於合成點雲後，模型性能的比較顯示，那些在合成資料上達到卓越精度的模型在大多數類別中的精度略低，使其更接近從真實資料獲得的結果。CenterPoint 和 TransFusion 的精度大幅提高，縮小了兩種資料之間的精度差距。這尤其可以在圖 4 中看到，其中可以找到每個模型在不同資料類型下的平均 AP。精度的損失並不意味著我們的處理沒有減少領域差距，相反，在大多數情況下，精度差距都減少了。由於應用於點雲的處理是有效的，使得合成點雲和真實點雲更加相似，因此真實領域和合成領域的 AP 差異減小了。

雖然所有模型在真實資料上累積點雲時都獲得了顯著的改進，但合成資料卻不總是如此，唯一表現相同的是 CenterPoint。PointPillars 和 SSN 幾乎沒有注意到單次掃描和累積點雲之間的變化。TransFusion 的準確度在累積點雲時降低，增加了領域差距。我們得出結論，在合成點雲的情況下，累積可能會根據所使用的模型產生不同的結果，因為它可能是有害的或有利的。

TransFusion 模型在不同領域的準確度差異顯著降低。在沒有點雲累積的情況下，我們的處理將差異減少了 36.4%，而在有累積的情況下，差異則為

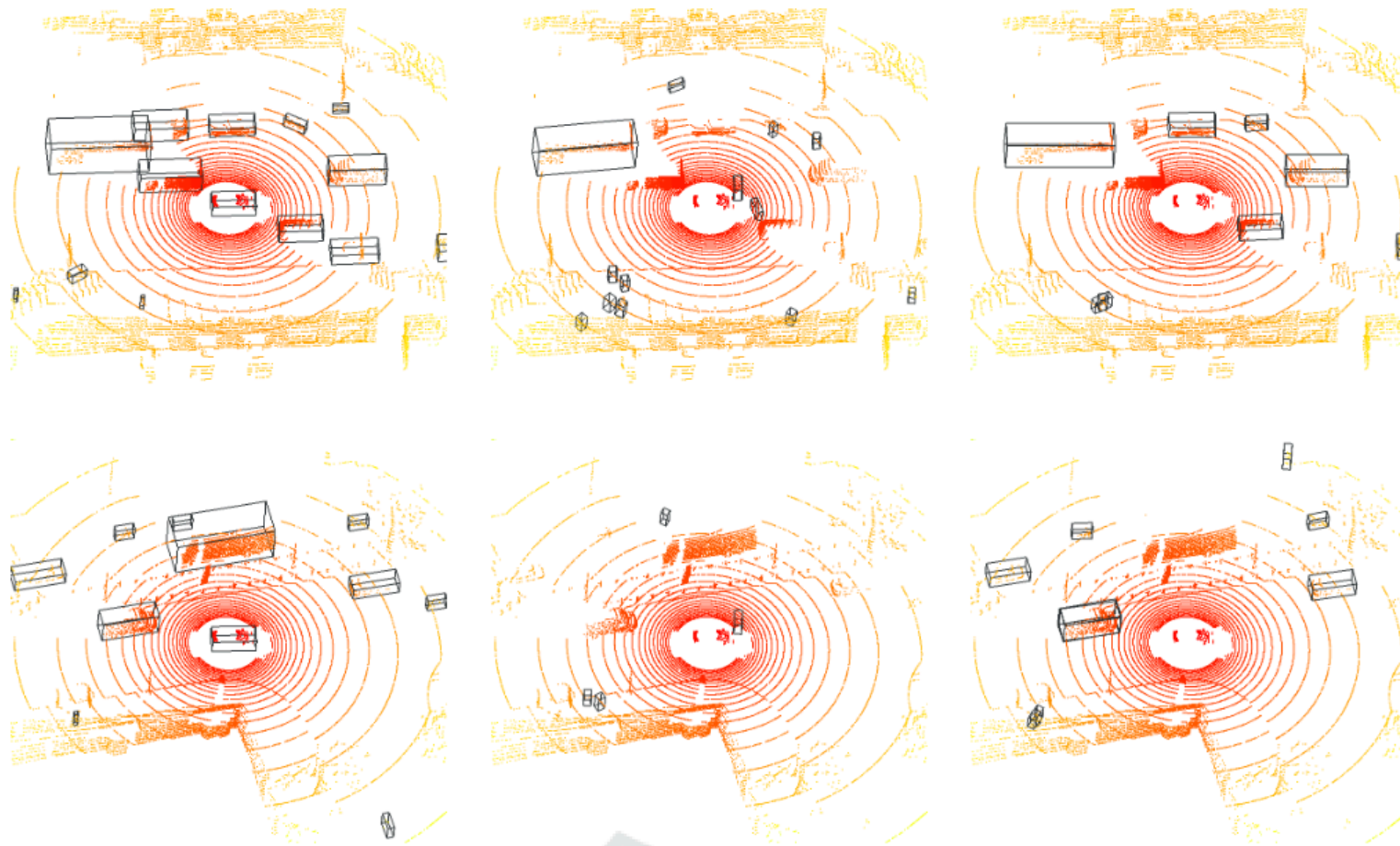


Figure 5: Comparison of detections in synthetic point clouds obtained with the CenterPoint model without point cloud accumulation. In each row, there is a frame of a different scene, the left column (a) contains the ground truth bounding boxes, the middle column (b) the detections without cloud processing and the right column (c) the detections with our cloud processing.

reduced by 48.6%. CenterPoint also reduces the accuracy difference with and without point cloud accumulation by 24.1% and 22.8%, respectively. With PointPillars, we have only achieved a reduction of 0.8% with point cloud accumulation and with SSN we have only archived a reduction of 0.7%. Our method effectively minimizes differences across all models, particularly in the car and pedestrian classes. The impact extends to other classes in most models. Based on these results, we conclude that our method reduces the difference between real and synthetic point clouds, even though high-performing models may not be as noticeably affected by our processing proposal.

In this paper, we also present qualitative results in Figure 5, illustrating the detection performance of our models on synthetic data and processed synthetic data in comparison to ground truth annotations. These images show the effectiveness of our proposed point cloud processing techniques. By comparing the model's detections on raw synthetic data to those on processed synthetic data, it is remarkable how our point cloud processing method significantly enhances the detection results, as a consequence of the domain gap reduction. These images highlight the improvements achieved by our approach, demonstrating the enhanced accuracy and robustness of the detection system when applied to processed point cloud data.

5 CONCLUSIONS AND FUTURE WORK

Recent advancements in driving simulators have transformed them into powerful tools for data generation. However, despite the easy access to data through these tools, there remains a question about their suitability for validating Automated Driving (AD) functions. Our study shows that there is a domain gap between real and simulated point cloud data.

Our study introduces an innovative synthetic data generation pipeline that creates standardized data adaptable to diverse sensor setups.

We study the evaluation of different LiDAR-based 3D object detection architectures concerning their performance with both real and synthetic data. Our findings reveal distinct responses across the models. PointPillars and SSN exhibited robustness to domain gap effects, while CenterPoint and TransFusion exhibited higher challenges in object detection when exposed to synthetic data. These observations emphasize the algorithmic influence on domain gap effects, underscoring the need for a prior analysis to assess how synthetic data may affect a specific algorithm.

Furthermore, we examined the domain gap between real and synthetic point clouds, identifying key distinctions. Our analysis revealed variations in reflected point numbers, structural differences in ring patterns, and fluctuations in point intensity values.

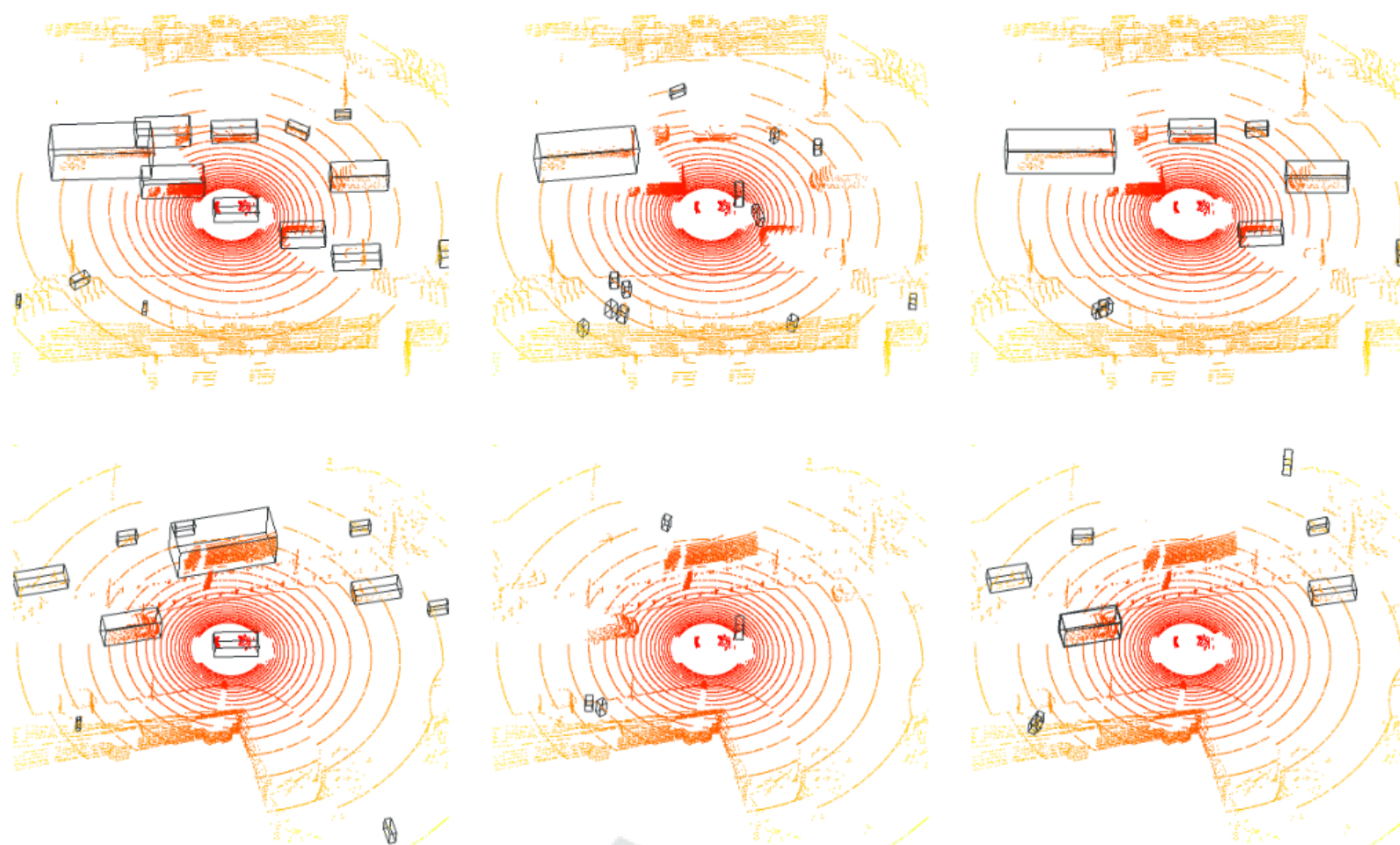


圖 5：使用 CenterPoint 模型在沒有點雲累積的情況下，合成點雲中檢測結果的比較。每一行都是不同場景的一幀，左欄 (a) 包含真實邊界框，中間欄 (b) 是沒有點雲處理的檢測結果，右欄 (c) 是我們點雲處理後的檢測結果。

減少了 48.6%。CenterPoint 也分別減少了 24.1% 和 22.8% 的點雲累積前後的準確度差異。使用 PointPillars，我們在點雲累積方面僅實現了 0.8% 的減少，而使用 SSN，我們僅實現了 0.7% 的減少。我們的方法有效地最小化了所有模型之間的差異，特別是在汽車和行人類別中。這種影響延伸到大多數模型中的其他類別。根據這些結果，我們得出結論，我們的方法減少了真實和合成點雲之間的差異，儘管高效能模型可能不會像我們的處理提案那樣明顯受到影響。在本文中，我們還在圖 5 中呈現了定性結果，說明了我們的模型在合成資料和經過處理的合成資料上的偵測效能，並與真實標註進行了比較。這些圖像顯示了我們提出的點雲處理技術的有效性。透過比較模型在原始合成資料和經過處理的合成資料上的偵測結果，我們的點雲處理方法顯著增強了偵測結果，這是由於領域差距縮小所致。這些圖像突顯了我們方法所實現的改進，證明了當應用於經過處理的點雲資料時，偵測系統的準確性和穩健性得到了增強。

5 結論與未來工作

駕駛模擬器在近期取得了進展，使其成為強大的資料生成工具。然而，儘管透過這些工具可以輕鬆取得資料，但它們是否適合驗證自動駕駛 (AD) 功能仍然存在疑問。我們的研究顯示，真實和模擬點雲資料之間存在領域差距。

我們的研究引入了一種創新的合成資料生成管線，可建立適用於各種感測器設定的標準化資料。

我們研究了不同基於光達的 3D 物件偵測架構在真實和合成資料上的效能評估。我們的研究結果揭示了模型之間的不同反應。PointPillars 和 SSN 對領域差距效應表現出穩健性，而 CenterPoint 和 TransFusion 在接觸合成資料時在物件偵測方面表現出更高的挑戰。這些觀察結果強調了演算法對領域差距效應的影響，突顯了需要事先分析以評估合成資料可能如何影響特定演算法。

此外，我們檢查了真實和合成點雲之間的領域差距，並確定了關鍵區別。我們的分析揭示了反射點數量的變化、環形圖案的結構差異以及點強度值的波動。

In response to these insights, we propose a domain gap reduction process for point clouds. This process proves its effectiveness through clear qualitative enhancements and a substantial reduction in accuracy gaps among various models when comparing their performance on real and synthetic data. Notably, this reduction is particularly prominent in the case of point cloud accumulation, where the CenterPoint and TransFusion models exhibit accuracy differences that are reduced by 22.8% and 48.6%, respectively. This approach can be applied to more reliably validate AD functions using synthetic point clouds.

This paper, has not investigated the potential of training models using solely synthetic data or in conjunction with real data, and it does not assess the contribution of these data in training, with or without processing. The exploration of this task is deferred to future work. Regarding the domain gap, although it has been possible to reduce it, it still exists and has not yet been completely reduced; continuing with the quantification and reduction of the domain gap in point clouds is still a pending and developing task.

ACKNOWLEDGEMENTS

This work has received funding from Basque Government under project AutoTrust of the program ELKARTEK-2023. This work is partially supported by the Ministerio de Ciencia, Innovación y Universidades, AEI, MCIN/AEI/10.13039/501100011033.

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為回應這些見解，我們提出了一種用於點雲的領域差距縮減流程。此流程透過清晰的定性增強以及在比較真實和合成資料上的效能時，顯著縮小了各種模型之間的準確度差距，證明了其有效性。值得注意的是，這種縮減在點雲累積的情況下尤為突出，其中 CenterPoint 和 TransFusion 模型的準確度差異分別減少了 22.8% 和 48.6%。這種方法可用於更可靠地使用合成點雲驗證 AD 功能。

本文尚未探討僅使用合成資料或結合真實資料來訓練模型的潛力，也未評估這些資料在訓練中（無論是否經過處理）的貢獻。這項任務的探索將延至未來的研究。關於領域差距，儘管已能縮小，但它仍然存在且尚未完全消除；繼續量化和縮小點雲中的領域差距仍是一項待處理且正在發展中的任務。

致謝

這項工作獲得巴斯克政府 ELKARTEK-2023 專案 AutoTrust 的資助。這項工作部分由西班牙科學、創新與大學部 (Ministerio de Ciencia, Innovación y Universidades)、AEI、MCIN/AEI/10.13039/501100011033 支持。

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